JÖNKÖPING UNIVERSITY
School of Engineering

Adaptive vs Plain-text AI Chatbot Outputs

Effects on Students' Perceived Clarity, Cognitive Load, and Format Preference

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JÖNKÖPING 2026 June

This final thesis has been carried out at the School of Engineering at Jönköping University within [see main field on previous page]. The authors are responsible for the presented opinions, conclusions and results.

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Scope:	15 hp (first-cycle education)
Date:	2026-06-04

Abstract

Most current AI chatbots default to Plain-text outputs with only slight formatting, such as headers and bullet points. Recent industry updates have begun to introduce more adaptive elements such as widgets and cards. The Cognitive Load Theory states that the structure of presented information directly affects the mental effort required to process it. There is no direct research on how Adaptive AI chatbot output formatting affects cognitive load compared to Plain-text. Therefore, the purpose of this study is to compare adaptive and Plain-text AI chatbot output formats in terms of how university students perceive their clarity and cognitive load, and to examine how their preferences for each format vary across three task types. A within-subject user study was conducted, with a total of 20 university student participants. The Adaptive vs Plain-text formats were tested on three different task types - Grammar checking, decision making, and exam preparation. Mixed methods were used to collect quantitative ratings for perceived clarity and cognitive load, and qualitative feedback. Custom-built prototypes were used for both conditions to control content and response time. Results showed Adaptive UI rated higher in perceived clarity and lower in cognitive load across all three task types, and participants preferred the Adaptive UI for all task types. These findings suggest that Cognitive Load Theory and multimedia learning principles extend to AI chatbot interfaces, where Plain-text outputs impose avoidable extraneous load, while adaptive formats reduce it.

Keywords: AI chatbots, adaptive user interfaces, cognitive load theory, perceived clarity, multimedia learning, human-computer interaction, university students, interface design

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1 Introduction

Artificial Intelligence (AI) tools like chatbots are becoming increasingly popular among students, who use them for everyday academic tasks ranging from quick fact lookups to complex questions (Ravšelj et al., 2025). As these tools grow more common in students' academic lives, the way they present information becomes a critical design concern.

While AI chatbots can be helpful for learning, most of these systems present information as long blocks of plain-text. In the context of this study, plain-text refers to the generic output of most current AI chatbots, such as ChatGPT, Copilot, and Gemini, which could include minor text formatting such as bullet points and headers, and does not include adaptive visual components such as cards or interactive widgets. This format is simple and universal; however, research in cognitive science suggests that visual presentation, not just content complexity, affects how easily information can be processed and understood (Mayer, 2009; Sweller, 1988). A recent development in this area is the emergence of adaptive User Interface (UI) outputs. These are chatbot responses that dynamically change their visual format based on the type of question asked, using elements such as cards, tables, and widgets. A recent example of this is Anthropic's Visual and interactive content update, which introduced dynamic visualisations tailored to the user's prompt (Anthropic, 2026). Despite these developments, little research has been done to look into how such formats compare to Plain-text outputs and how this difference affects students' experience with AI chatbots. This study addresses that gap.

1.1 Problem statement

The way information is presented to users has been an important topic of interest in human-computer interaction and interface design, because the format of information directly affects how well it can be understood (Mayer, 2009). The Cognitive Load Theory (CLT) states that visual structure and layout affect the mental effort needed to process information regardless of how complex the content is (Sweller, 1988; Mayer, 2009). Research on AI-driven information presentation suggests that adaptively structuring how content is visually structured can reduce cognitive load during decision-making tasks (Wang et al., 2026).

Even with these findings, Adaptive UI outputs in AI chatbots remain largely unexplored in research. While tools like Anthropic’s Claude have begun introducing dynamic response formats, such as interactive maps and data visualisations that are tailored to the user's prompt (Anthropic, 2026), little research has looked into how formats like these compare to Plain-text outputs in terms of cognitive load and user experience. Without fully understanding these effects, there is a risk of deploying powerful AI tools whose default output formats unintentionally overload students or fail to support efficient information processing. Examining how response format affects students’ experience across different academic task types could provide

valuable insights into how AI chatbot interfaces should be designed to better support students.

1.2 Purpose and research questions

Drawing on the problem statement, it is evident that most AI chatbots widely used by students default to Plain-text outputs. Further, it is evident that despite the emergence of adaptive response interfaces, there is little evidence on how these formats affect students' experience compared to Plain-text outputs.

Consequently, the purpose of this study is:
To compare adaptive and plain-text AI chatbot output formats in terms of how university students perceive their clarity and cognitive load, and to examine how their preferences for each format vary across three task types.

In order to accurately fulfil the purpose of this study, two research questions have been defined.

Cognitive Load Theory suggests that how information is visually presented directly affects the mental effort required to process it (Sweller, 1988). Students interacting with AI chatbots mainly encounter Plain-text outputs, and little is known about how these formats differ from adaptive outputs in perceived clarity and cognitive load. These two are treated as related but distinct subjective measures. CLT provides the link between them: when a format reduces extraneous load, less mental effort is needed to process the information, while also making the presentation feel clearer. Clarity reflects how the format feels to read, while mental effort reflects how hard it is to extract information from it. Measuring both gives a full picture of how response format affects students' interaction with these prototypes.

Hence, the study's first research question is:
[1] How do Adaptive chatbot output interfaces differ from Plain-text outputs in perceived clarity and cognitive load among university students?

Students use AI chatbots for a variety of academic task types, and preferences for response format may vary depending on the type of task. Understanding these preferences is important for designing AI outputs that best support diverse academic needs. Hence, the study's second research question is:
[2] What are university students' preferences towards Adaptive chatbot output interfaces compared to Plain-text across different academic task types?

1.3 Scope and limitations

This study examined the effect of UI format in academic AI chatbot use. It is limited to self-reported subjective measures, and does not cover objective performance like task accuracy or retention of information. The study was conducted with a sample of 20 university students, which limits the generalisability of findings to other

populations, such as professionals or non-academic users in general. Additionally, the study captures single interaction sessions per participant, and long-term effects such as changes in preferences over time are not measured. The use of pre-generated static responses is necessary to control for answer quality, though the findings may not fully reflect the variability of real-time AI-generated outputs.

1.4 Disposition

The rest of this thesis is organised as follows. Chapter 2 presents the theoretical framework, introducing the main theories and concepts that support the study. Chapter 3 describes the method and implementation, which outlines the research approach, data collection, data analysis, and considerations. Chapter 4 presents the results of the study. Chapter 5 discusses the findings in relation to the research questions, existing literature, and methodological choices. Chapter 6 presents the conclusions of the study as well as future research suggestions.

2 Theoretical framework

This study is mainly grounded in the Cognitive Load Theory, which provides a base to examine how UI format affects the mental effort required to process AI Chatbot outputs. Research on multimedia learning principles and AI-driven information presentation provides the applied context for this study. These frameworks together provide understanding for how the visual presentation of information affects mental effort and motivate the investigation of perceived clarity and format preference as related outcomes.

2.1 Cognitive Load Theory

Cognitive Load Theory (CLT), introduced by Sweller (1988), proposes that human working memory has a limited capacity and that learning and comprehension are directly affected by the mental effort required to process information. CLT was selected for this thesis as it directly addresses the role of visual presentation in mental effort, which is a key aspect of this study's focus on UI format.

CLT (Sweller, 1988; Duran et al., 2022) states that the cognitive effort required to process information depends not only on the complexity of the subject (intrinsic load) but also on the way information is visually presented (extraneous load). Extraneous load is not caused by the specific content, unlike intrinsic load, and can be reduced through better UI design (Sweller, 1988; Mayer, 2009). Reducing extraneous load frees up cognitive resources for germane load, which is the effort the brain uses to understand and retain information (Sweller, 1988; Duran et al., 2022). This means that poorly structured information, redundant text, and the need to mentally search for scattered details - all increase extraneous load, whereas well-formatted presentations can reduce unnecessary cognitive effort (Duran et al., 2022).

In the context of the study, UI format is treated as a potential source of extraneous load. RQ1 directly investigates whether adaptive chatbot output interfaces differ from plain-text in the cognitive load and perceived clarity experienced by students, which is directly grounded in CLT. Since UI format functions as a source of extraneous load, a well-structured and adaptive interface should improve how clearly users believe the information is presented and therefore reduce the mental effort required to extract it. Perceived clarity here is treated as a subjective judgement of how clearly information is perceived to be presented within a specific format.

2.2 Multimedia Learning Principles in AI Chatbot Interfaces

Adaptive UI refers to interfaces that dynamically adjust their output format based on the prompt or the needs of the user. Modern AI chatbots such as ChatGPT, Claude, and Perplexity already employ slight forms of adaptive formatting, such as switching between Plain-text, bullet points, tables, and code blocks depending on context. However, there is still an absence of actual research that explicitly looks at how

adaptive output formatting influences cognitive load in comparison to Plain-text responses.

While CLT explains why visual presentation affects mental effort, Mayer's (2009) Cognitive Theory of Multimedia Learning extends this into specific design principles for how information should be visually structured to reduce extraneous cognitive load. The signalling principle (Mayer, 2009) states that highlighting essential information through visual cues helps users locate and organise relevant information, which lowers the cognitive resources spent on filtering. The split-attention principle (Chandler & Sweller, 1992; Mayer, 2009) states that physically integrating related information into a unified visual structure reduces the extraneous load caused by mentally connecting separated elements. Together with related design principles such as progressive disclosure (Nielsen, 2006) and Gestalt grouping (Harley, 2020), these principles describe how visual structure can directly reduce extraneous cognitive load and guide attention more effectively.

Recent empirical research has begun to apply these principles in AI-driven contexts. Wang et al. (2026) examined how AI-restructured review presentation formats (bullet-point versus paragraph) affected consumers' cognitive load during decision-making. Using a 2x2 between-subjects design with 240 participants, they found that bullet-point formatting reduced cognitive load by 11.8% in high cognitive demand conditions, while no major difference emerged under low cognitive demand conditions. Though the context differs from AI chatbot outputs, these findings support the claim that AI-driven format restructuring can reduce cognitive load, while also suggesting that the size of this effect depends on the complexity of the underlying task. This gap between the principles established in multimedia learning, the early empirical findings in the adjacent AI context, and the lack of direct evidence in AI chat interfaces forms the basis for this study.

Together, these theoretical perspectives imply that Adaptive UI layout may reduce extraneous cognitive load, which would lower mental effort and increase clarity. However, whether this effect holds equally across different task types and whether it translates into format preference remains untested in AI chatbot interfaces, forming the basis of this study's two research questions.

3 Method and implementation

This chapter describes and motivates the methodological decisions made throughout the study. It begins by outlining the overall research design and approach, followed by a description of how the data was collected and analysed. The chapter concludes with an account of how validity and reliability were ensured, as well as relevant ethical and scientific considerations.

3.1 Research Design and Approach

3.1.1 Study Design

As the study aimed to examine the effect of two different AI output User Interfaces on cognitive load, perceived clarity, and format preference, a controlled within-subjects user study was deemed most appropriate. As within-subjects designs offer greater statistical power and control for individual differences between participants, this makes them especially suited for comparative studies (Charness et al., 2012).

By keeping the content of AI responses constant and changing only the UI (Plain-text vs Adaptive), the study isolates UI as the independent variable, with cognitive load, perceived clarity, and format preference being the dependent variables. The full set of prompts and corresponding outputs for each task type is presented in Appendices 1-12.

The independent variable in this study is not a single visual feature, but rather the Adaptive UI as a whole design unit that combines things like signalling, progressive disclosure, and common region grouping. These principles support each other, and to reflect how adaptive design works in practice, they were not isolated individually. Therefore, RQ1 examines how the Adaptive UI as a whole compares with the Plain-text, rather than measuring the effect of a single principle.

The study used a mixed-methods approach - combining quantitative measures of cognitive load and clarity with qualitative user feedback. Quantitative measures are efficient and validated (e.g., the Paas scale; Paas, 1992), and produce comparable data, such as means and distributions, which are essential for result statistics. These quantitative measures alone cannot capture the reasons why participants may find one format clearer or more preferable than the other, which is why it is necessary to also collect qualitative measures. Qualitative feedback allows participants to explain specifically which elements were a distraction or not, providing information that numerical scales cannot. As neither a quantitative nor a qualitative approach by itself is sufficient to fully understand the research problem, a mixed methods design was adopted (Creswell, 2014).

The targets for this study are university students, as they are frequent users of AI tools and have been the commonly studied population in prior research on adaptive AI learning systems, cognitive load, and academic outcomes.

3.1.2 Prototype Development

Both the adaptive and Plain-text prototypes were developed as single-page web applications. The application was designed to match real-world chatbot websites – inspiration for the development was mostly taken from ChatGPT and Claude, featuring a sidebar navigation, prompt box and other minor elements. This study aimed to replicate real-world AI chatbot tools in order to simulate a realistic interaction experience without exposing participants to a specific tool they may already have strong familiarity with. Full control over the application allowed for the mitigation of additional confounding variables - response generation time was identical across both conditions, and by using a custom-built interface rather than an existing platform, any prior bias toward a specific tool's design (e.g., recognising ChatGPT's UI) was avoided. Figures 1 and 2 below illustrate the contrast between the two conditions for the Exam Preparation task: the Plain-text prototype shown in Figure 1 and the corresponding Adaptive prototype shown in Figure 2. Equivalent prototype pairs for each task type are presented in Appendices 1-12.

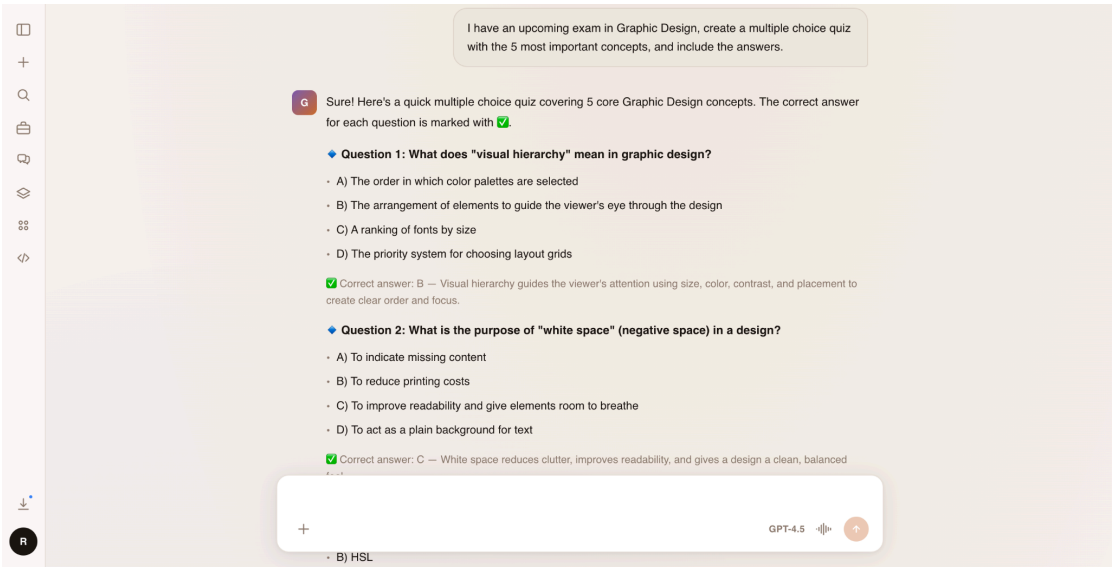


Figure 1: *Prototype of the Plain-text exam preparation task, presenting information as a Plain-text AI chatbot in a Plain-text manner.*

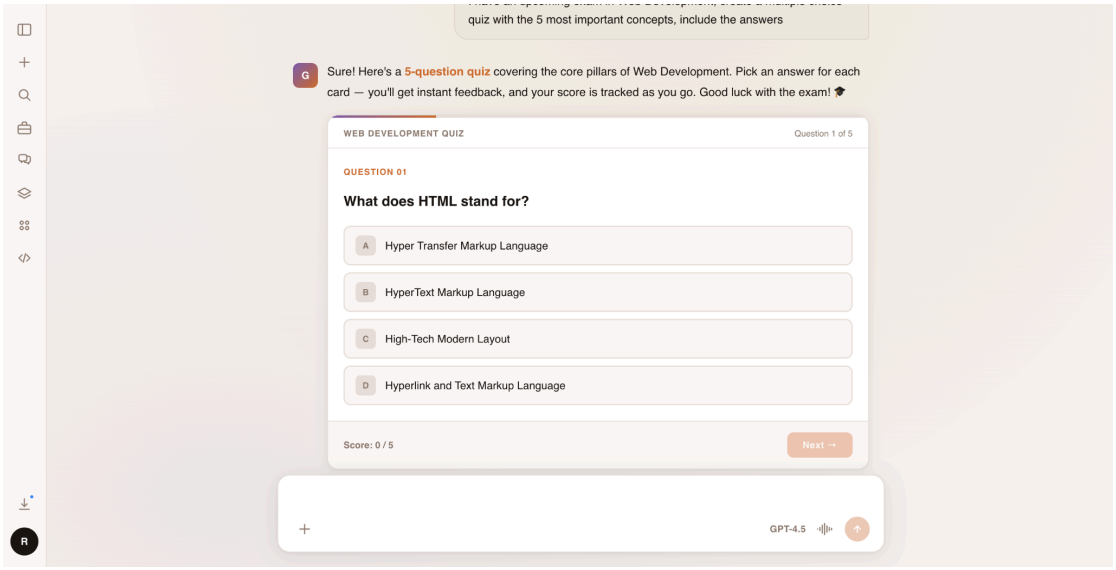


Figure 2: *Prototype of the adaptive exam preparation task, presenting information in an adaptive way, based on the user's needs.*

A key methodological principle is controlling confounds related to "answer quality" by generating all responses using the same model, GPT-5.3 Instant (OpenAI, 2026). This model was selected for the study period; it is the default model in ChatGPT, meaning it reflects real-world use encounters. The model selector shown in the prototype screenshots (Figures 1-2 and Appendices 1-12) displays a placeholder label and does not reflect the model used.

It was an important part of this study to keep the content similar so that any observed difference was due to the UI format rather than the difference in the content. A consequence of the control could be seen in the exam preparation task, as the correct answer for each question had to be present in both conditions. Though for the Plain-text version, this meant the answers would be visible as users were looking at the questions. While in the adaptive version, the answers were hidden using progressive disclosure. Removing it from the Plain-text version would have introduced differences in the content itself and therefore undermined the study's ability to isolate format as the independent variable. The difference between the two conditions is consequently not the presence of the answer, but rather the concealment of it.

Adaptive UI

Following the multimedia learning principles outlined in Section 2.2, the Adaptive UI prototype was designed using the following principles, each affecting specific design decisions across the three task types. For each task type, the AI output grouped relevant information into specific self-contained visual components (widgets) instead of continuous text. The use of widgets in this case aligns with the Gestalt principle of common region, as related information was grouped within a visual boundary, so

users see it as a unified unit, and reduce the need to mentally assemble scattered content (Harley, 2020).

As for the Grammar check task specifically, highlighting the changes (added, modified, removed) to the text directs the attention to the relevant information without having the users read the full text, which reflects Mayer's (2009) signalling principle. Chandler and Sweller (1992) talk about the split-attention effect, which states that physical integration of related elements reduces extraneous cognitive load. Therefore, in the Decision-making task, the pros and cons were placed in adjacent columns so users do not need to mentally integrate information from separate locations.
For the Exam task, users were only exposed to one question at a time, to keep their focus on one thing and not be distracted by the other questions, following the principle of progressive disclosure to reduce cognitive overload (Nielsen, 2006). This can also be seen in the Decision-making task, where the UI adapts to the user's input and only shows the relevant information at once; in this case, users only see one pros and cons list at a time.

The interactive elements, such as navigation between questions or sections, also contribute to the adaptiveness of the interface. Rather than delivering a fixed paragraph or page, interactivity allows the UI to respond to the user's actions and adjust based on what the user chooses to engage with.

Plain-text UI
The Plain-text UI replicated the default output formats of current generative AIs. Responses were delivered as a continuous text, with minor formatting such as bold text, bullet points, and headers, which is consistent with how current AI chatbots present their responses by default. No interactive elements, widgets, or visual groupings were included. This format serves as the baseline condition and represents the most common chatbot outputs users currently interact with.

3.1.3 Counterbalancing and Sample Size

In order to mitigate carryover effects, the experiment order was counterbalanced - half of the participants first encountered the Plain-text version, half the adaptive version. Similarly, the sequence of task types (Decision Making, Exam Preparation, and Grammar Check) varied across participants, ensuring that neither UI condition nor task familiarity could influence the results. Counterbalancing controls the order of conditions so that it is no longer a confounding variable, meaning that any difference between conditions cannot have been caused by order alone (Cuttler, 2017).

As a between-subjects design would require a much larger group of participants and risk skewing results due to differences between participants, rather than the effect of UI format itself, the study used a within-subjects design, due to each participant contributing data for both conditions, producing 60 paired data points per measure. A

sample of 20 student participants was selected as a feasible minimum for a within-subjects descriptive study at a bachelor's thesis scope. This balances recruitment effort against the goal, which is to detect consistent patterns across conditions (Adaptive vs Plain-text) and task types. This sample size supports descriptive comparison and identification of qualitative themes, though it is not intended for inferential generalisation to a wider population.

3.2 Data collection

Each task was designed to reflect a realistic student interaction with an AI chatbot - participants were asked to engage with the output as they naturally would, scanning content, reviewing comparisons, or checking corrections. The goal was not to absorb the information itself, but to experience how it was presented and structured.

The three academic task types that were selected for this study are grammar checking (Appendices 1-3), exam preparation (Appendices 9-12), and Decision-making (Appendices 4-8). These task types were chosen because they represent some of the most common ways students use AI chatbots academically. Grammar checking and editing are common student AI use cases (Yeung et al., 2025). Exam preparation and assessment-related AI use have grown significantly, with a survey of over 1,000 university students finding that 88% reported using generative AI for assessments, including studying and concept explanation (Freeman, 2025). Decision-making tasks require students to gather and weigh information and form a reasoned position, a process which is linked to the "finding evidence" and "understanding topics" patterns identified in student AI use (Black & Tomlinson, 2025). These three task types were also chosen to ensure variety, in the nature of the task as well as the UI elements each response would require.

Cognitive load in experimental research is commonly measured using subjective self-report scales. The NASA-TLX (Hart & Staveland, 1988) is a widely used multidimensional scale covering six subscales, including mental demand, effort, and frustration, but its complexity makes it better suited to multi-component tasks. The Paas (1992) scale is a single-item 9-point rating of perceived mental effort, validated extensively in instructional research and better suited to single-task experiments. For this reason, the Paas scale was selected as the measure of cognitive load in this study. Participants were asked to rate their perceived mental effort on a 9-point scale ranging from 1 (very, very low mental effort) to 9 (very, very high mental effort) (see Appendix 13). This scale was selected due to its simplicity, low response load, and its established validity in experimental settings (Paas, 1992). Perceived clarity was also measured using a 9-point scale for consistency. Single-item rating scales have been shown to be a valid alternative to multi-item scales for measuring perceived usability in studies involving simple and concrete user interfaces (Gräve & Buchner, 2024).

During each study session, we verbally administered the questions to the participant and recorded their responses in a Google Form. The full form layout used during data collection can be seen in Appendix 13. Participants were initially briefed about the study and requested to sign a consent form as part of a standardised procedure. They then received an introduction to the experiment, which briefly outlined the process. Prior to the main tasks, in order to establish baseline context, participants were asked questions, such as AI chatbot usage frequency (rated on a 1-5 scale, where 1 = never and 5 = daily), the types of questions they typically ask, and which AI chatbots they normally use.

The main phase consisted of three task types, each completed under both UI prototypes. The participants were not required to fully complete the tasks or come to a final conclusion; they were told to interact with the AI output as they normally would, for example, by reviewing the grammar changes, going over a few of the exam questions mentally or by interacting with the interface, or going over the decision-making output. This was intended to recreate real-world AI interactivity without introducing task performance pressure as a variable.

For each task, the participant interacted with one UI version first, then responded to a set of post-task questions - including perceived clarity of information presentation (1-9 scale, where 1 = very unclear and 9 = very clear), whether any elements drew attention away from relevant content (open-ended), and perceived mental effort required to extract the information (Paas, 1-9 scale). Perceived clarity of information presentation was included as a dependent variable, as the visual structure of an interface directly affects extraneous cognitive load, and unclear presentation increases the unnecessary mental effort demanded of the user (Sweller, 1988).

The participant then completed the same type of task using the alternate UI version and answered the same post-task questions. Once both UI versions had been completed for a given task type, participants were asked two closing questions: whether they had previously used AI for that type of task, and which of the two versions they would prefer to use for it. This process was repeated across all three task types. Additionally, at the end of the main phase, they were given the opportunity to state any final notes or remarks.

The post-task questions were designed to directly address the study's research questions. Perceived clarity (1-9 scale) and mental effort (Paas, 1-9 scale) directly target RQ1 by capturing cognitive load and clarity differences between the two UIs. The open-ended question regarding distracting elements was specifically included to investigate whether the additional visual components of the Adaptive UI could become a source of distraction rather than an aid. The closing questions per task type directly target RQ2 by capturing student preferences towards each UI format across

different academic task types. Together, these questions provide both the comparability and depth needed to fully address both research questions.

3.3 Data analysis

Quantitative data were initially summarised using Google Forms' built-in response overview, providing a distribution of responses across conditions. The descriptive statistics were calculated, which included means, standard deviations, and medians. This was calculated in Google Sheets for perceived clarity and cognitive load ratings across both UI conditions and all three task types. For each of the two dependent variables, statistics were calculated separately for each condition and task combination, alongside aggregate values across all three task types for each UI condition. Means and medians were reported together so that the closeness between them could be used as an informal check on whether results were driven by outliers. Scores were then compared across the Adaptive and Plain-text UI conditions to identify differences in participants' scores. The analysis followed a set procedure - responses were first fully read through to gain familiarity with the content, and afterwards sorted by UI condition and task type. Responses that contained only a "No" answer were set aside, and substantive responses were tagged with short descriptive codes capturing the type of distraction reported (e.g., scrolling between sections, exam prep answers visible). Codes that recurred across multiple participants were grouped into possible themes. Recurring themes were then identified to detect any trends in participants' responses and interaction with each UI format. Comments that did not fit the main themes but added nuance were retained and reported separately rather than put into a theme.

3.4 Validity and reliability

Internal validity was further supported by counterbalancing both condition and task order across participants, as described in Section 3.1, minimising the influence of carryover and familiarity effects on results. To further reduce memory effects across the two interface conditions, each task type uses paired prompts that are structurally equivalent but differ in surface details, such as using a different paragraph for the grammar check or a different comparison for the decision-making task. This ensures participants cannot simply reuse the previously seen answer while still performing the same kind of information extraction. Using structurally equivalent but distinct prompts across different conditions also means that any differences in cognitive load scores reflect the UI format, rather than differences in task difficulty between prompts.

Reliability was supported through the use of the Paas scale - a validated research instrument that we used across all participants to ensure that cognitive load was measured in the same way.

All sessions were conducted in person on a single device provided by us, which ensured a consistent environment across all participants. We were present throughout each session to answer questions and ensure the protocol was being followed correctly. Each session lasted approximately 15 minutes, which was considered sufficient to complete all tasks while minimising fatigue effects, which could otherwise influence cognitive load ratings.

3.5 Considerations

Participants were informed of the study's purpose and their right to withdraw at any point. In order to protect participant privacy, no personal information was gathered, and all of the answers were kept anonymous.

Although the study provides insight into how students interact with AI output forms, the sample size and similar participant profiles limit how far the results can be generalised. To reach more general conclusions, larger and more varied sample sizes would be required in future research.

4 Results

This section is structured as follows: the first part presents participant profile results, covering participants' reported daily chatbot use, which chatbots they use, and the types of questions they normally ask. The second part consists of results for RQ1 addressing perceived clarity and cognitive load, as well as qualitative findings on distraction. The third part is about RQ2, presenting format preference across task types, and the section concludes with a short summary of all the results.

4.1 Participant profile results

The sample size consists of 20 university students, as previously mentioned in Section 3.1.3. Most participants are frequent AI users - specifically, 9 participants reported daily use (5/5), 5 reported near daily (4/5), 5 reported moderate use (3/5), and 1 reported infrequent use (2/5), with the mean being 4.10. None of the participants reported “never” having used AI, meaning all of the findings come from participants with established AI familiarity.

Most participants reported using multiple chatbots, the most common one being ChatGPT (mentioned by 14 participants), Claude (9 participants), Gemini (8 participants) and Perplexity (7 participants). As participants had been exposed to a range of AI chatbots and chatbot interfaces, the results are not shaped by familiarity with just one tool.

Collected data shows that the most frequent types of questions participants usually ask are summarised in Table 1, with Paraphrase/Grammar (mentioned by 17 participants), Decision making (13 participants), Exam preparation (13 participants), Fact lookup/check (10 participants), and Explanation (7 participants). This overlaps with the three task types selected for the study, supporting the choice of tasks as relevant.

Question Type	**n**
Paraphrase / Grammar	17
Decision Making	13
Exam Preparation	13
Fact lookup/check	10
Explanation	7

Table 1: *Frequency of reported question types participants reported asking AI chatbots.*

4.2 Perceived clarity and cognitive load (RQ1)

This section presents the results for Research Question 1: how the Adaptive and Plain-text UI differ in perceived clarity and cognitive load. Both measures were rated by participants on a 9-point scale across the three task types: Decision Making, Grammar Checking, and Exam Preparation. The study followed a within-subjects design, meaning all 20 participants completed tasks under both conditions. Descriptive statistics are presented first, followed by qualitative findings on distraction.

4.2.1 Descriptive statistics

Descriptive statistics for perceived clarity and perceived mental effort (Paas) were calculated for each UI condition, both collected across all three task types and broken down per task type. Across all tasks combined, the Adaptive UI was rated as substantially clearer than the Plain-text UI (M = 8.43, SD = 0.46 vs M = 5.73, SD = 0.93 on a 9-point scale), and also required considerably less mental effort for the students to extract information (M = 2.12, SD = 0.83 vs M = 4.92, SD = 1.09). The mean differences across the two conditions were +2.70 points for clarity and -2.80 points for Paas, indicating a consistent pattern in favour of the Adaptive UI on both measures.

When broken down by task type specifically, the same pattern can be seen across all three tasks, although the size of the difference varied. The positive clarity difference of the Adaptive UI was largest in the Grammar Check task (+4.35 points), and smaller, though still large, in the Decision Making (+1.95) and Exam Preparation (+1.80) tasks. A similar pattern was observed for mental effort, where the Grammar Check task showed the largest decrease in Cognitive Load scores under the Adaptive condition (-4.55 points), compared to the Decision Making (-1.85) and Exam Preparation (-2.00) tasks. The median values were close to the means in all cases, suggesting the results were not heavily influenced by outliers. The complete descriptive statistics are presented in the following table.

Task	Measure	Adaptive M (SD)	Adaptive Mdn	Plain-text M (SD)	Plain-text Mdn	Mean Diff (A–P)
Decision Making	Clarity	8.30 (0.66)	8.0	6.35 (1.42)	6.0	+1.95
Decision Making	Cognitive Load	2.20 (0.95)	2.0	4.05 (1.57)	4.0	-1.85
Exam Preparation	Clarity	8.45 (0.76)	9.0	6.65 (0.81)	7.0	+1.80

Exam Preparation	Cognitive Load	2.15 (1.50)	2.0	4.15 (1.53)	4.0	-2.00
Grammar Check	Clarity	8.55 (0.83)	9.0	4.20 (2.33)	3.0	+4.35
Grammar Check	Cognitive Load	2.00 (0.92)	2.0	6.55 (1.90)	7.0	-4.55
Aggregate	Clarity	8.43 (0.46)	-	5.73 (0.93)	-	+2.70
Aggregate	Cognitive Load	2.12 (0.83)	-	4.92 (1.09)	-	-2.80

Table 2: *Descriptive statistics for perceived clarity and cognitive load by UI condition and task type (n = 20). Both measures are reported on 9-point scales, where higher clarity scores indicate clearer presentation and higher cognitive load scores indicate greater mental effort.*

The same pattern is visualised across task types in Figures 3 and 4, where Figure 3 displays the mean perceived clarity rating, and Figure 4 displays the mean Paas scores. Both figures show the Adaptive UI outperforming the Plain-text UI on every task, with Grammar Check showing the largest gap on both measures.

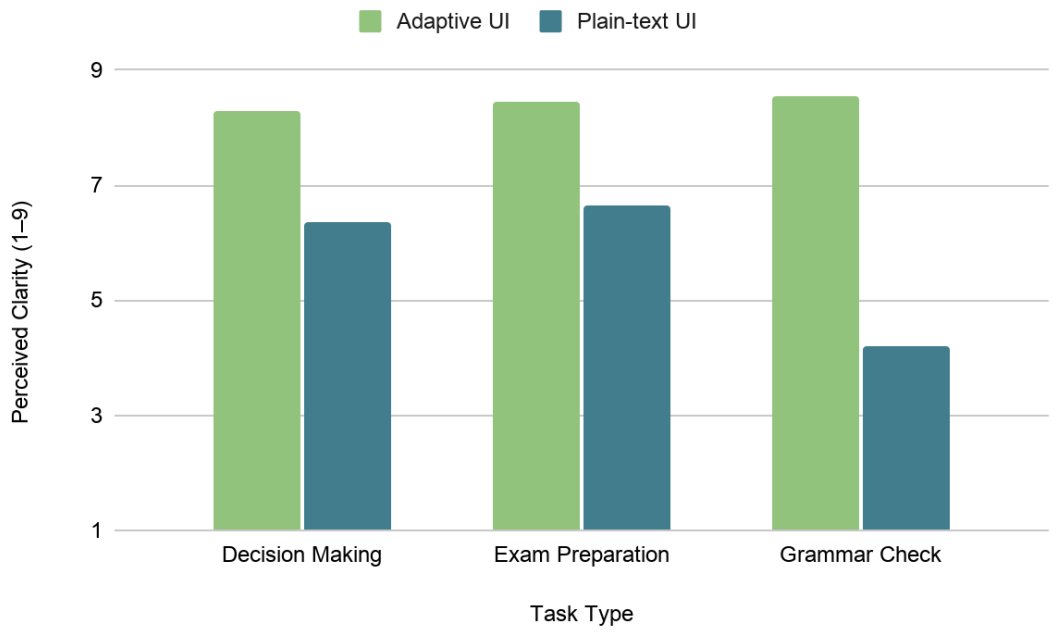


Figure 3: *Mean perceived clarity ratings by UI condition and task type (n = 20).*

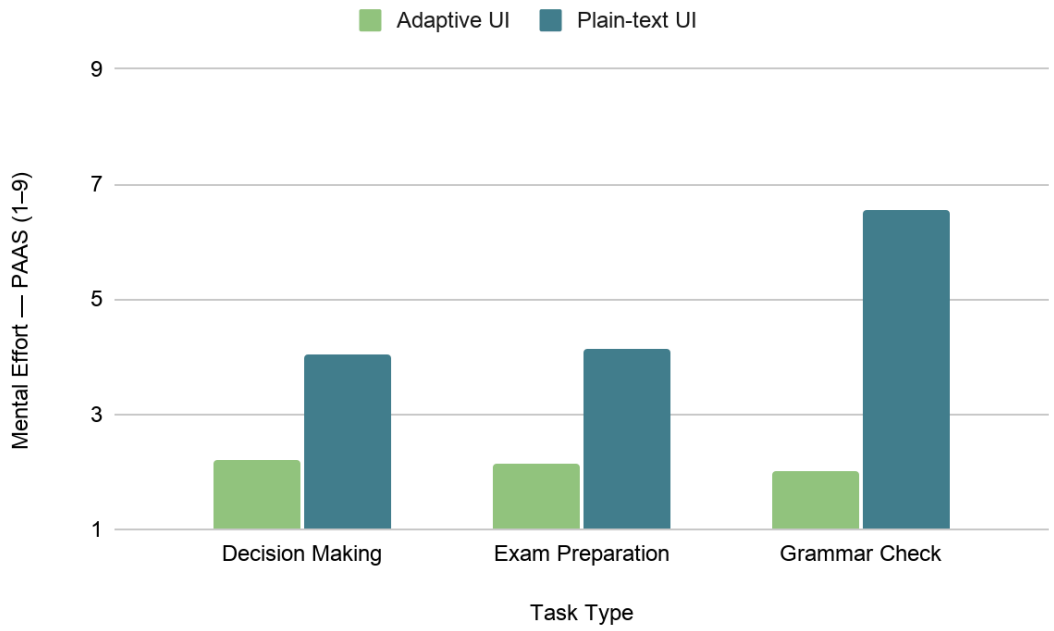


Figure 4: *Mean Paas scores (perceived mental effort) by UI condition and task type (n = 20).*

4.2.2 *Qualitative findings on distraction*

Out of the 60 responses collected per condition (20 participants x 3 tasks), 27 of them contained feedback for the Plain-text UI, while only one contained feedback for the Adaptive UI, displaying a large difference. This pattern reflects the cognitive load and clarity findings, as participants were more distracted by elements of the Plain-text UI.

The themes identified in each condition are presented below.

Themes from Plain-text UI

The analysis of the Plain-text UI responses about distraction revealed 3 key themes: *navigation difficulty*, *dense and unstructured presentation*, as well as *exposure of answers* in the exam preparation task.

Theme 1, referring to navigation between content received 9 mentions. Participants frequently mentioned having to scroll/jump back and forth between different sections to connect related information. This was mentioned most in the Decision Making and Grammar Check tasks. One participant commented, “Comparing the two, you had to go up and down to match the pros and cons”(P4, Decision Making). Another noted, “It takes more effort to check what has been changed, as you need to go back and forth” (P6, Grammar Check). Another commented, “You can't tell what's changed and have to constantly go back and forth” (P14, Grammar Check).

Theme 2, referring to dense information and unstructured presentation received 6 mentions. Multiple participants mentioned that the Plain-text UI was visually dense or not well structured, making it hard to locate relevant or specific information, hence causing back and forth scrolling as mentioned in Theme 1. One participant noted it was “too text-based”, and “it’s just listing information, it’s a bit distracting” (P7, Decision Making). Another commented that “all the information was on top of each other, which was a little distracting” (P16, Decision Making). Others noted how it was “just a big paragraph” and for the Grammar check “you weren't able to see what had been changed”.

Theme 3, referring to exposure of answers in Exam Task received 9 mentions. A theme that emerged specifically in the Exam preparation task was that the Plain-text UI displayed the exam questions and answers together. Participants specifically mentioned this as a distraction from the intended outcome when using AI to study. One participant explained, “When you want to learn with AI, if you have the answer in front of you at the same time as the question, your eyes go to the correct answer, and you don't think by yourself" (P12, Exam Preparation). Another commented that “It’s hard to focus on one question because the other questions were right there, and I kept looking at the answer right below" (P2, Exam Preparation). Others mentioned how it was hard not to cheat, as the answer was right there, highlighted with a green check mark.

Distraction comments on Adaptive UI

Of the 60 distraction responses collected for the Adaptive UI, only one contained substantive content beyond a “No” answer. The participant noted that the visual highlighting elements in the Adaptive UI, while purposeful, slightly drew their focus towards the specific changes, “the different colour and highlights - there was an intention behind it. It didn’t fully draw me away, but I did spend more time looking at what was changed instead of the whole text” (P8, Grammar Check).

Summary of qualitative findings

Feedback on the Plain-text UI focused on structural problems with the format itself, like effortful navigation, dense unstructured presentations, and content visibility that affected participants' intended tasks. One participant reported that the additional components in the Adaptive UI distracted them from the underlying content. These findings directly address the concern raised in the methods chapter - whether richer interfaces could become a distraction rather than an aid - which, in this study, they did not.

4.2.3 Additional Notes and Observations

In addition to the additional notes question asked at the end of the user study, observations and participant comments not captured by the survey were recorded as well. These notes were grouped into two major themes.

Theme 1: Interface Interactivity and layout

Two participants did not understand the interactive elements for the Adaptive UI, specifically for the Decision-making task and the Exam preparation task. Another commented, “I wish the Adaptive Decision-making task were stacked on top of each other rather than next to each other; the horizontal layout makes it harder to follow” (P13, Additional Notes).

Theme 2: Clarity and Structure

One participant noted the Adaptive UI was much clearer and easier as all of the information was not in one place, making it easier to search and extract what they needed. Another participant explicitly said, “The adaptive was much clearer, even the fact that they were in widgets just makes everything more put together and easier for my brain to understand” (P19, Additional Notes). A third participant noted they didn't expect there to be something as good as the adaptive version, as they have only seen the standard Plain-text ones.

Additionally, one participant stated a noteworthy observation regarding the Grammar checking task. They commented that while they preferred the Adaptive UI as it highlighted all the changes, they believed that the Plain-text version may be more beneficial for learning grammar more deeply, as it requires more active learning and engagement with the text.

4.3 Format preference across task types (RQ2)

This section addresses RQ2, more specifically the participant format preference across the three task types.

Overall, the Adaptive UI was largely preferred over the Plain-text UI across all three task types, as shown in Figure 5. Grammar check showed complete preference towards the Adaptive UI (20/20), followed by Exam Preparation, with one participant indicating a preference for a combination of both formats (19/20), and Decision Making, with three participants preferring the Plain-text UI (17/20).

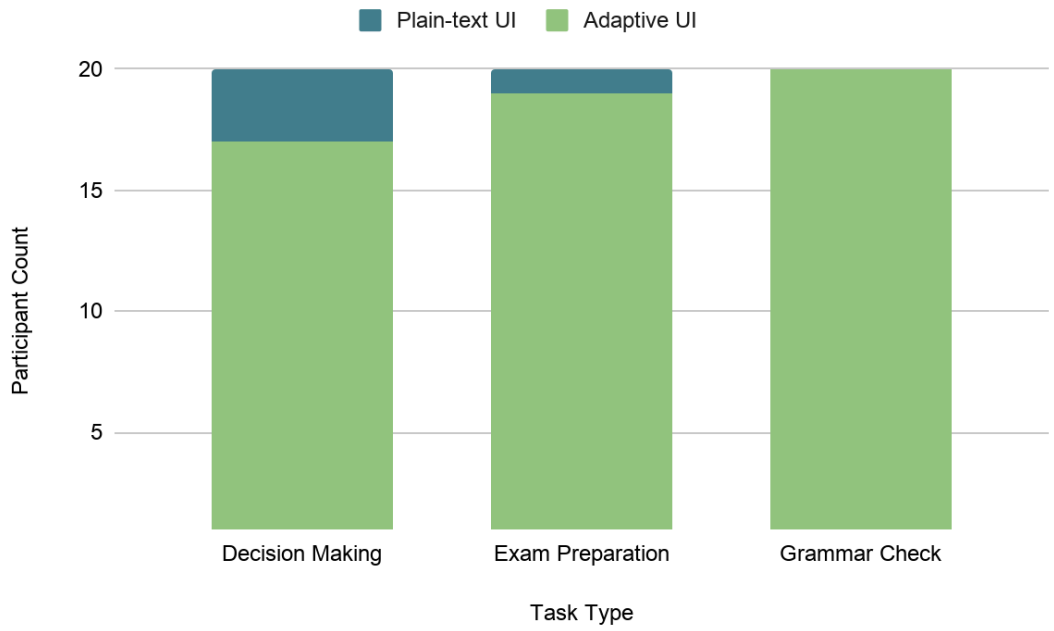


Figure 5: *Distribution of UI format preferences across task types (n = 20).*

Results display a strong preference towards the Adaptive UI across all task types, with Grammar Check showing the strongest preference (20/20) and Decision Making the weakest (17/20).

These findings are discussed in the following discussion chapter, in relation to the CLT and previous research on adaptive AI interfaces, analysing the observed patterns and what they imply.

5 Discussion

This chapter discusses the findings and results of the study. Starting with Section 5.1, which discusses the results in relation to the two research questions and existing literature, followed by Section 5.2, which discusses methodological strengths, limitations, improvements, and to what degree the research questions were answered.

5.1 Result discussion

This study aimed to compare how adaptive chatbot response interfaces and Plain-text outputs are experienced by university students across different academic task types. This study is guided by two research questions:
1. How do Adaptive chatbot output interfaces differ from Plain-text outputs in perceived clarity and cognitive load among university students?
2. What are university students' preferences towards Adaptive chatbot output interfaces compared to Plain-text across different academic task types?
The following findings are discussed in relation to the Cognitive Load Theory and prior research on adaptive interfaces.

Participant profile influence
All participants had prior experience using AI chatbots, with the sample skewing towards frequent AI users (mean 4.10/5), meaning the findings reflect how experienced AI users respond to the two formats rather than the broader sample of university students. Whether a less-experienced user would show the same pattern is unknown. Most participants also reported using multiple chatbots regularly (ChatGPT, Claude, Gemini, Perplexity) - this multi-chatbot exposure means participants' reactions were not shaped by a familiarity with one specific tool, but rather reflect comparison against the category of AI chatbots they know. The strong preference for Adaptive UI is notable given the sample composition - despite participants' habit of being used to Plain-text formats from their regular use, they preferred the unfamiliar Adaptive format. This strengthens the interpretation that preference was driven by format effectiveness and not by familiarity bias.

RQ1 - Clarity and cognitive load findings
The main finding for clarity and cognitive load is that the Adaptive UI is clearer and requires lower mental effort across all three tasks. This is in line with the Cognitive Load Theory, which states that a better structured presentation reduces cognitive load. As the content of the responses was constant between each condition - intrinsic load remained the same, the difference comes from extraneous load (UI). The Adaptive UI's structure - presenting the chatbot's output into cards, interactive widgets, and inline highlighting - as expected, has reduced extraneous cognitive load. Lower extraneous load means more working memory was available for germane processing, which is consistent with participants reporting both less effort and better clarity. The Adaptive UI was deliberately prototyped using the multimedia learning principles

introduced in Section 2.2 and applied in Section 3.1.2. The reduction in cognitive load that was perceived in the Adaptive UI across all task types suggests that these principles functioned as theory would predict, with the structure of the visual output appearing to reduce extraneous load the way that Cognitive Load Theory suggests.

Based on the participants' responses, Grammar Check showed the largest effects; this is due to the imbalance between low intrinsic load and high extraneous load in the Plain-text version. The task itself is simple, as participants were asked to interact with the output as they normally would - by checking what was changed, added or removed, but finding those changes in a Plain-text paragraph requires intensive visual scanning, which imposes high extraneous load. On the other hand, Adaptive UI's inline highlighting eliminated almost all of that extraneous load effort. This reflects the signalling principle (Mayer, 2009), which states that visual cues highlighting essential material reduce the cognitive resources spent on locating relevant information. Both Decision Making and Exam Preparation tasks showed similar but smaller effects due to the plain-text format not being as actively negative to these tasks. The Adaptive UI's enhancements still applied the multimedia learning principles, such as placing comparable information side by side, reflecting the split-attention principle, and presenting one exam question at a time, reflecting progressive disclosure (Mayer, 2009; Nielsen, 2006). The reductions were smaller because the plain-text imposed less extraneous load to begin with. This pattern is consistent with the CLT - the more extraneous load a format imposes, the more room there is for a better structured format to reduce this load. Grammar Check showed the largest gap, as the Plain-text UI version imposed the highest extraneous load, leaving the most room for improvement.

These findings are broadly consistent with Wang et al. (2026), who found that the effect of format on cognitive load was not uniform, but rather dependent on task conditions. A similar pattern is observed in this study, in which the size of the format effect varied across task types.

RQ1 - Qualitative findings on distraction

As reported in Section 4.2.2, the distraction comments were heavily concentrated in the Plain-text condition, with only one meaningful Adaptive UI response, which noted that the visual highlights in the Grammar Check task drew attention to the specific changes rather than the actual content, indicating that, though there was a minor distraction, the visual signalling principle (Mayer, 2009) worked as intended. This also addresses the explicitly raised concern in Section 3.2: whether Adaptive UI might add visual complexity, therefore adding distraction. As the data show, visual richness does not correspond to distraction. This aligns with Cognitive Load Theory's prediction, which states that it is unstructured information that increases extraneous load, not the number of visual elements. The data support the interpretation that visual

complexity, when applied with design intent (per prototype design rationale in Section 3.1.2), reduces extraneous load rather than adds to it.

Unlike the Adaptive UI, the Plain-text version findings show distinct distraction themes, the most notable being answer exposure in the Exam Preparation task. As established in Section 3.1.2, the answer exposure in the Exam preparation task was a consequence of keeping content constant across conditions. The Plain-text version displays answers alongside the questions, while the Adaptive UI hides them until the user submits a response (progressive disclosure). Therefore, the findings show that the Adaptive UI's structure aligns more closely with the task's underlying goal of active recall.

For the Grammar Check task, the goal is scanning, so the adaptive format wins because it makes the differences more easily visible. For decision-making, the goal is comparative evaluation; in this case, the adaptive format wins due to its clear structure. For Exam Preparation, the goal is active recall, adaptive format wins again because it supports the process, and the Plain-text format actively undermines it by providing the correct answer right below each question. Together, these findings suggest that distraction in this study was caused by structural weaknesses of the Plain-text format, such as effortful navigation, dense text presentation and, in the case of Exam Preparation, misalignment with the task's underlying goal. No reported distraction was caused by the Adaptive UI's components.

The two RQ1 measures (clarity, cognitive load) capture perceived experience, meaning they don't directly measure objective cognitive load, comprehension, or attention. Participants reported lower mental effort with the Adaptive UI version, though this reflects how the format felt to them rather than how much cognitive work was actually performed. A user might still feel that a format is easier even if objective measures such as response time, error rate, and recall accuracy show no actual difference. This was acknowledged early on in the study, but due to the study's time constraints and our expertise, the Paas scale was still chosen as a measurement of cognitive load due to its validation in instructional research as a reasonable proxy.

RQ2 - Preference findings

Participants showed a strong preference for all three task types for the Adaptive UI with Decision Making 17/20 (85%), Exam Preparation 19/20 (95%, with one participant preferring a combination), and Grammar Check 20/20 (100%), suggesting the preference for Adaptive UI is not task-bound. While intuitive, it is still worth noting that participants who experienced an interface as clear and less effortful also preferred it - preference strength matches the size of clarity and cognitive load effects from RQ1. Taken together, the RQ2 findings show that participant preference was scaled with the size of the format effect on perceived experience. The task with the largest reduction in cognitive load for Adaptive UI (Grammar Check) also showed the

strongest preference, while the task with the smallest reduction (Decision Making) showed the weakest. This alignment between RQ1 and RQ2 suggests that preference followed perceived clarity and effort, rather than being shaped by aesthetics or novelty.

Theoretical contribution

This study extends Cognitive Load Theory and multimedia learning research into a new area of AI chatbot interfaces. Wang et al. (2026) demonstrated this effect in consumer decision contexts using a single format change (bullet-point versus paragraph). This study extends this finding into AI chatbot use with a richer Adaptive UI involving widgets, interactivity, inline highlighting, and progressive disclosure, and ultimately shows that the relationship between format and cognitive load holds in this context as well. Mayer's (2009) signalling and split-attention principles, originally developed for educational multimedia, also appear to function as predicted when applied to AI chatbot output. This is particularly relevant given current industry developments, such as Anthropic's Claude visual and interactive content update (Anthropic, 2026).

This study provides clear data for such design choices and demonstrates that the default Plain-text outputs impose avoidable extraneous load on users compared to adaptive formats. While these findings come from a limited sample, they offer a good demonstration that adapting the UI based on the task reduces perceived cognitive load in AI chatbots.

5.2 Method discussion

Strengths

This study has several strengths that support the validity of the results. The prompts used for each task were structurally equivalent but distinct, which controlled memory and carryover between conditions. Additionally, by having all sessions be in-person and on a single device, this allowed for a controlled lab setup and a consistent environment. The short (±15 min) session length was controlled for mental fatigue confounds. Furthermore, the prototype used was developed to keep variables such as AI output and response time constant - ensuring a consistent and controlled setting. The task types chosen for this study were grounded in actual student AI use. Finally, the strong results - mean and median values are closely aligned, meaning results are not driven by outliers.

Limitations

Despite these strengths, there are several limitations of this study that are worth noting. As mentioned before, due to the time constraint and our expertise, the results were descriptive-only; no inferential statistics were run; therefore, the patterns can only be described, and no statistical significance claims can be made. Even though the custom prototype allowed for consistency, it lacked real-tool affordances such as

follow up prompting. Additionally, since the Adaptive UI was unfamiliar to participants, it could have potentially increased preference scores via the novelty effect, which is the tendency for people to respond more positively to something just because it is new or unfamiliar. As already noted, the sample skews towards frequent AI users, with a mean of 4.10/5, making it hard to generalise these results to novice AI users. Finally, the question about AI use frequency had unlabeled midpoints, making the middle ratings hard to interpret.

Methodological improvements

Looking back, several methodological decisions could have been made differently or in addition. The first thing is running inferential tests, to complement the descriptive statistics - this would have allowed us to see the significance of the results. Secondly, objective measures such as time on task and accuracy would have allowed for comparison between perceived effort and actual task performance. Piloting the questionnaire first would have surfaced the issue with unlabeled middle points on the frequency scale, before the main data collection. Another improvement would have been the preference question, which used a forced binary choice (Adaptive or Plain-text) that captured the direction of preference but not the strength - for example, using a 9-point scale would have shown how strongly each format was preferred and allowed it to be compared more directly against clarity and cognitive load measures. Lastly, a larger sample size and a multi-session design would have allowed for more generalisable findings and data on long-term result effects.

Validity and reliability

Internal validity was well supported - the within-subjects design allowed for controlled individual differences between participants. Counterbalancing of both conditions (Adaptive-first vs Plain-text-first) as well as task - controlled carryover and order effects. Paired prompt structure allowed for reduced memory transfer between conditions while keeping the task difficulty comparable. Using the same model (GPT-5.3 Instant) allowed for identical content across conditions, isolating UI as the only varying factor (ensuring a constant intrinsic load). As previously mentioned in strengths, the controlled environment and pre-generated responses allowed for a controlled setting and removed AI output variability as a confound.

External validity is quite limited due to factors such as a sample size of 20 participants and all from a single university, sample skewing towards frequent AI users, limiting the generalisation. Single-session design does not provide insight into long-term preferences, and by only testing three types of tasks, other tasks, such as fact lookup or explanation, are left untested. Validity of the results measures was ensured with the Paas scale and a single-item clarity measure. Both of these scales are subjective, and though they present known limitations such as social desirability, they are validated and well-established. Reliability was ensured by participants using the same device, the same protocol and the same prototype. As previously mentioned, use of the Paas

scale ensured consistent measurement of cognitive load; this consistency can also be seen in the results - both the mean and the median are closely aligned across measures, as well as the consistent pattern across all task types. The sample size of 20 is sufficient for descriptive within-subjects analysis, but limits statistical reliability and generalisation.

These methodological reflections inform both the conclusion drawn from the study and the suggested further research presented in Chapter 6.

6 Conclusions and further research

This chapter presents the conclusions drawn from the study's findings and outlines directions for further research. Section 6.1 summarises the conclusions regarding the problem statement and discusses the study's practical and theoretical implications. Section 6.2. Suggests directions for future research.

6.1 Conclusions

Prior research has shown that restructuring information format can reduce cognitive load in consumer decision contexts (Wang et al., 2026), and multimedia learning principles such as signalling and split-attention have long been established as effective in reducing extraneous load (Mayer, 2009). However, there is a lack of research focusing on Adaptive UI chatbots specifically, which this study addresses. In response to RQ1, the Adaptive UI was perceived as clearer and less cognitively effortful than the Plain-text UI across all three task types. In response to RQ2, students preferred the Adaptive UI in every task type. These results indicate that the effect of output format on cognitive load observed in prior research extends into AI chatbot output, and this effect translates into a consistent preference for adaptive output formats over Plain-text ones.

Current AI chatbots default to Plain-text outputs, though research on Cognitive Load Theory suggests this may not be optimal. CLT states that the way information is visually presented directly affects the mental effort required to process it. This means that Plain-text formats could impose unnecessary extraneous cognitive load on users (Sweller, 1988; Mayer, 2009). More Adaptive UI in AI chatbots is emerging, with Claude's recent Visual and interactive content update, though the effects of these formats remain unstudied. Without understanding these effects, there is a risk of deploying tools whose default formats overload students. The findings of this study substantiate the concern raised in the problem statement: default Plain-text formats do appear to impose avoidable extraneous load on students, and Adaptive UI alternatives consistently produce clearer, less effortful, and more preferred experiences across the task types tested.

6.2 Further research

While this study addresses important questions relating to Adaptive UI in AI chatbot outputs, there are several things that remain open for further research. Firstly, replicating the study with inferential statistics on a larger sample would allow for more patterns to be observed and strengthen the generalizability of the study. Studying objective measures such as time on task and accuracy could also be valuable, and provide behavioural evidence in addition to the self-reported findings.

Future research could also explore multi-sessions and longitudinal designs to test whether preference persists or fades over time, addressing the potential novelty effect. Finally, broader tasks and content coverage, more complex interactions with the AI, using real-world tools rather than just custom prototypes, and testing on non-student participants would help establish the generalisability of these findings. The findings of this study offer an early but consistent pattern: Adaptive formats in AI chatbots present a clear positive effect on students' cognitive load and clarity.

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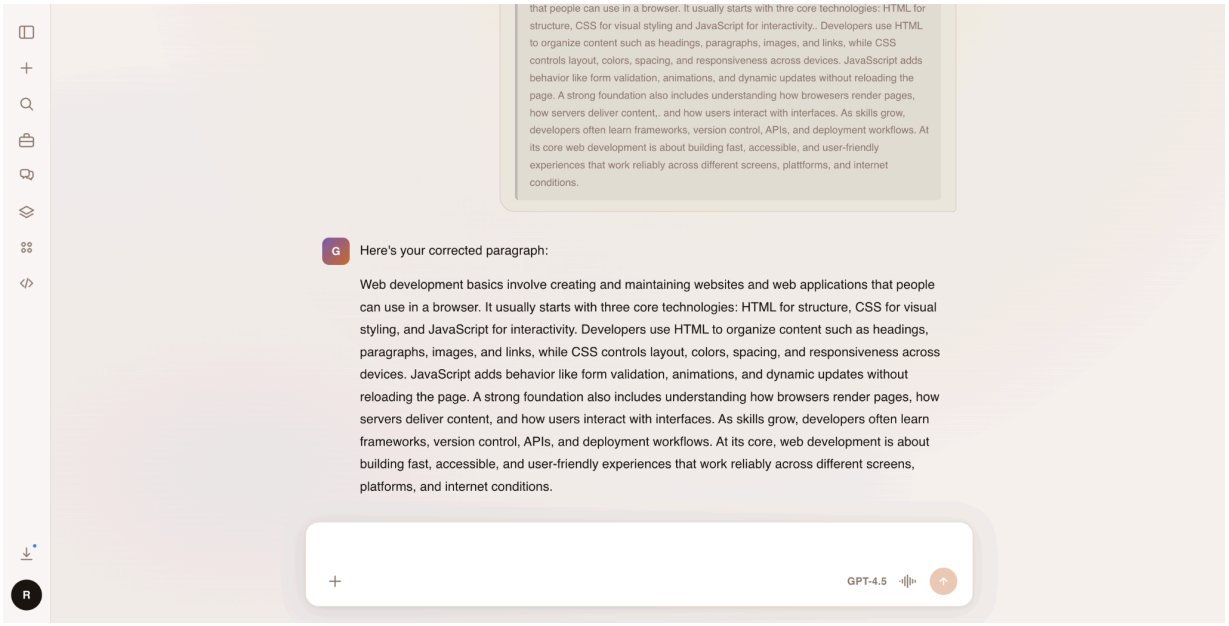
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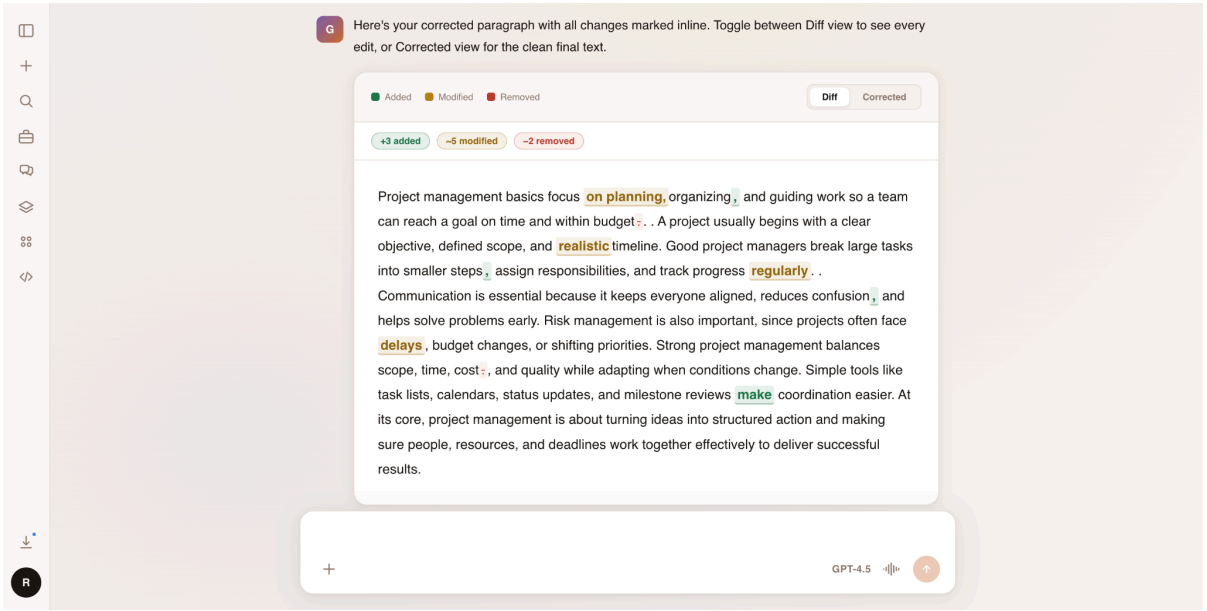
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8 Appendixes

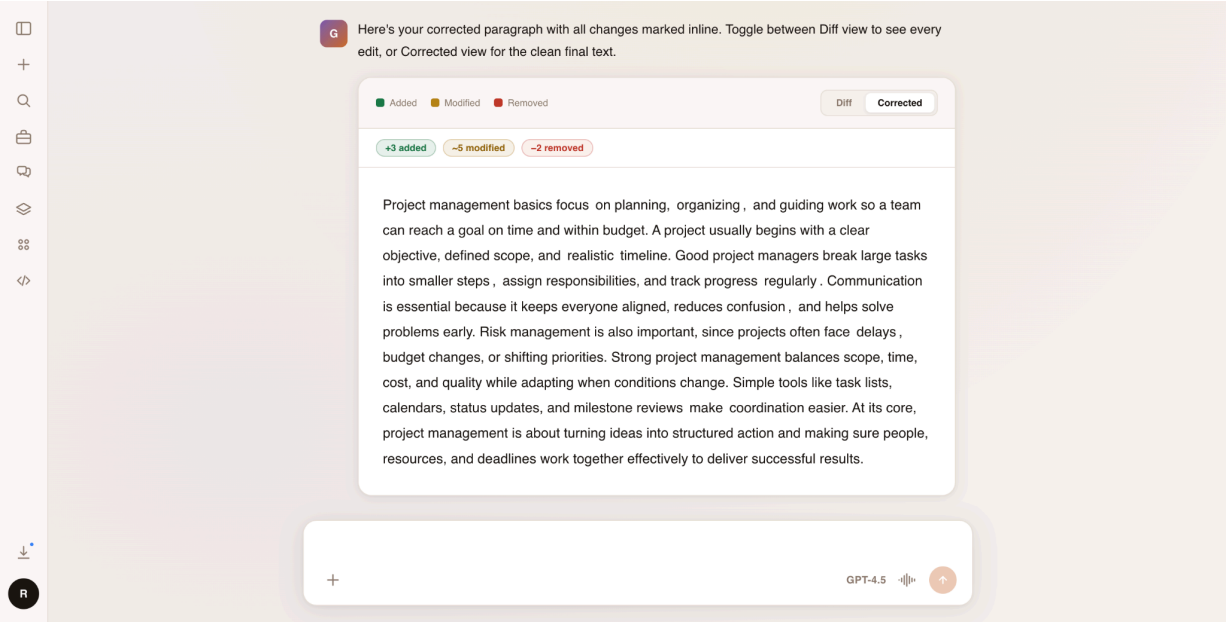
Appendix 1 Plain-text prototype of the grammar check task



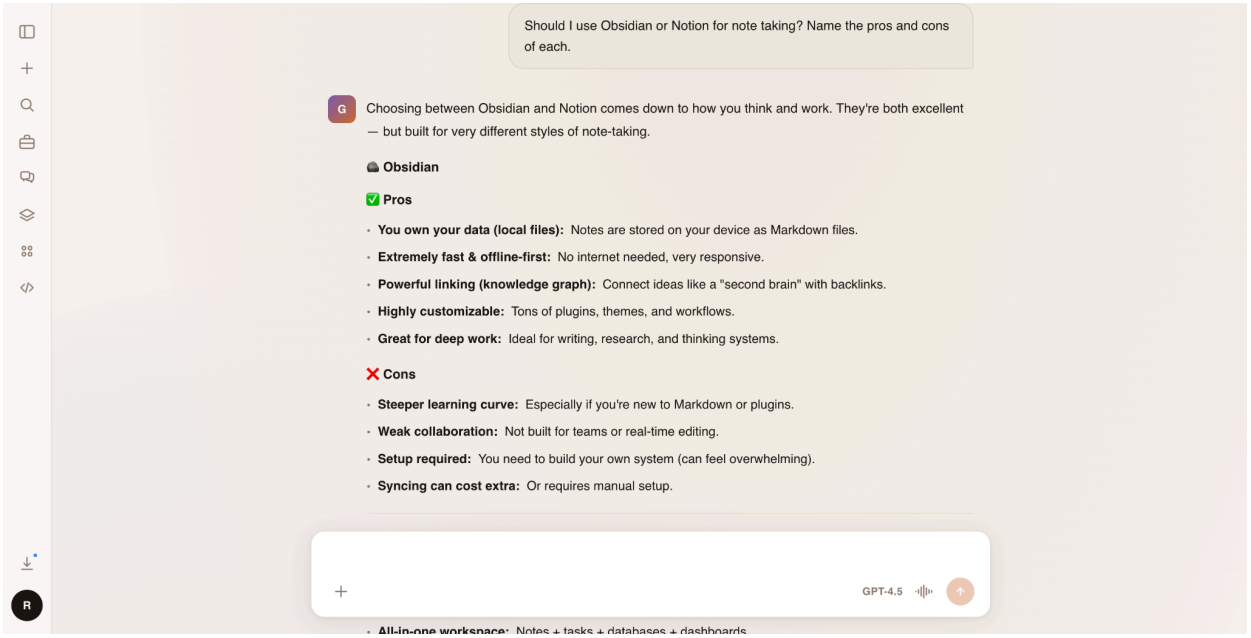
Appendix 2 Adaptive prototype of the grammar check task, Diff view



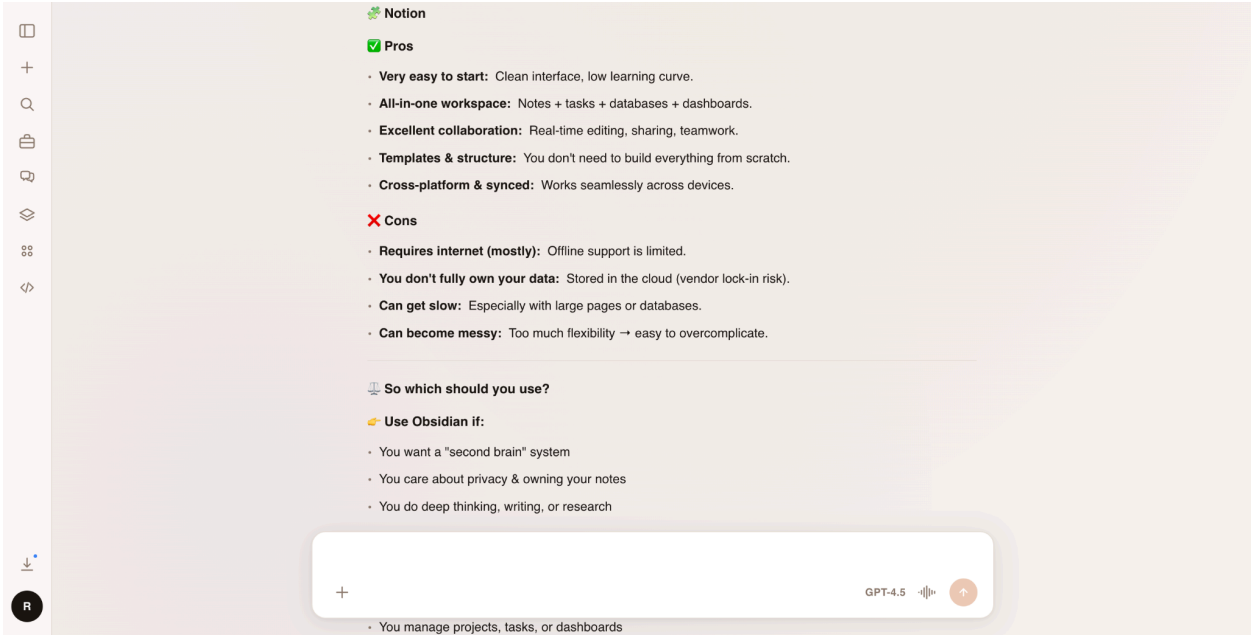
Appendix 3 Adaptive prototype of the grammar check task, Corrected view



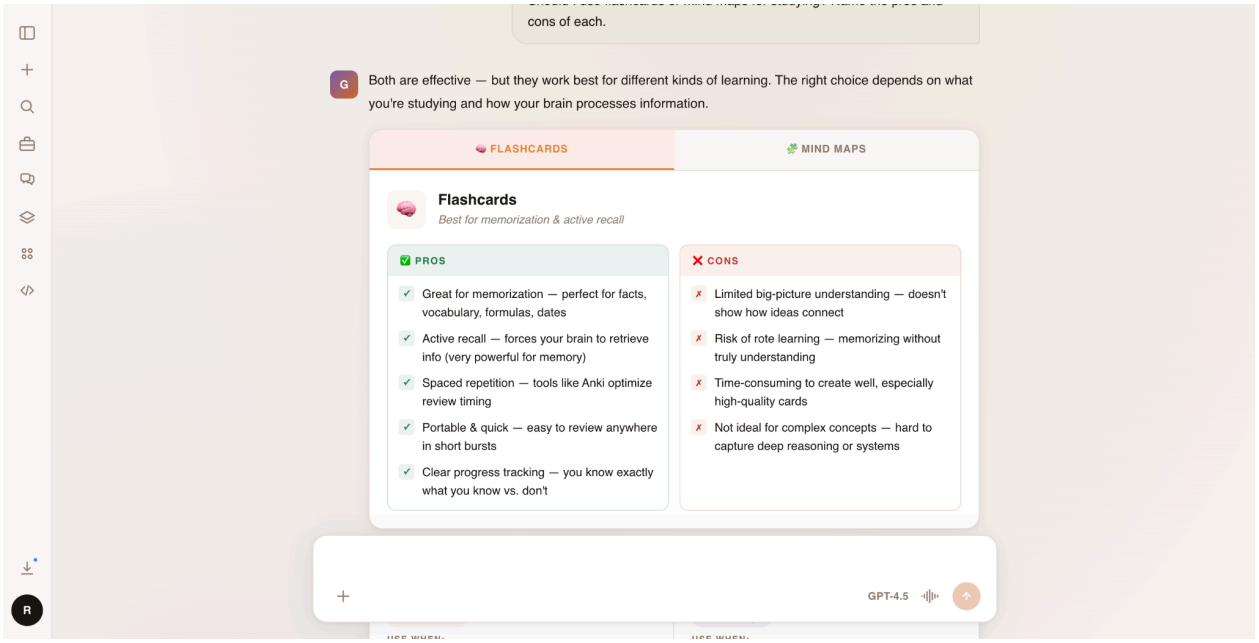
Appendix 4 Plain-text prototype of the Decision making task



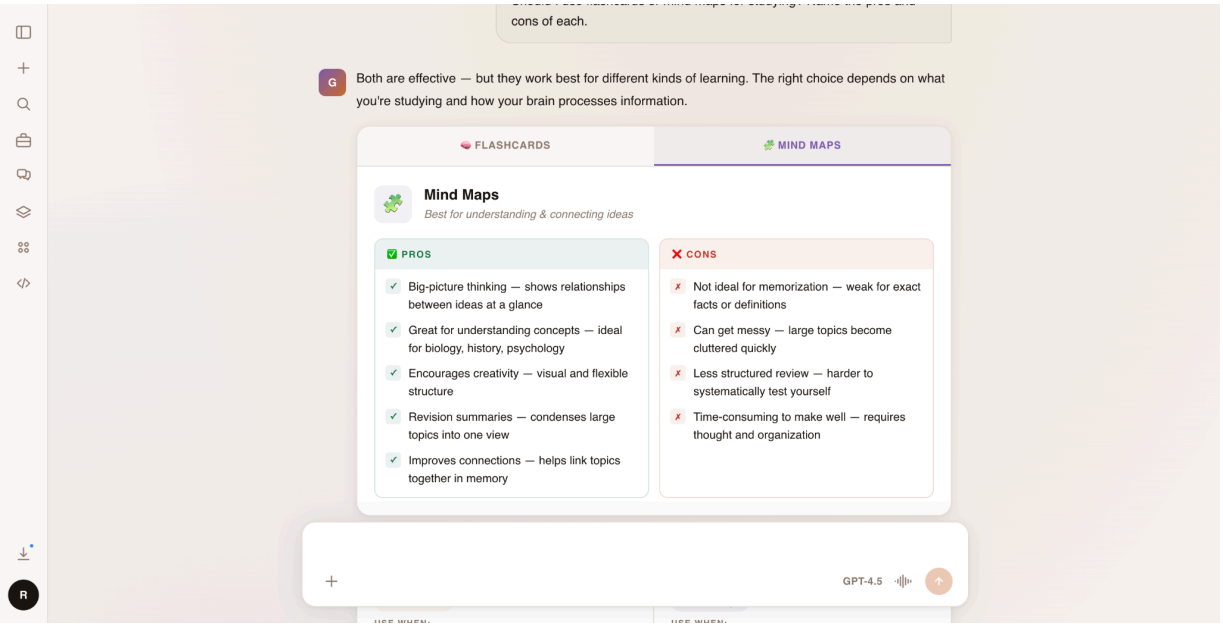
Appendix 5 Plain-text prototype of the Decision making task, conclusion section



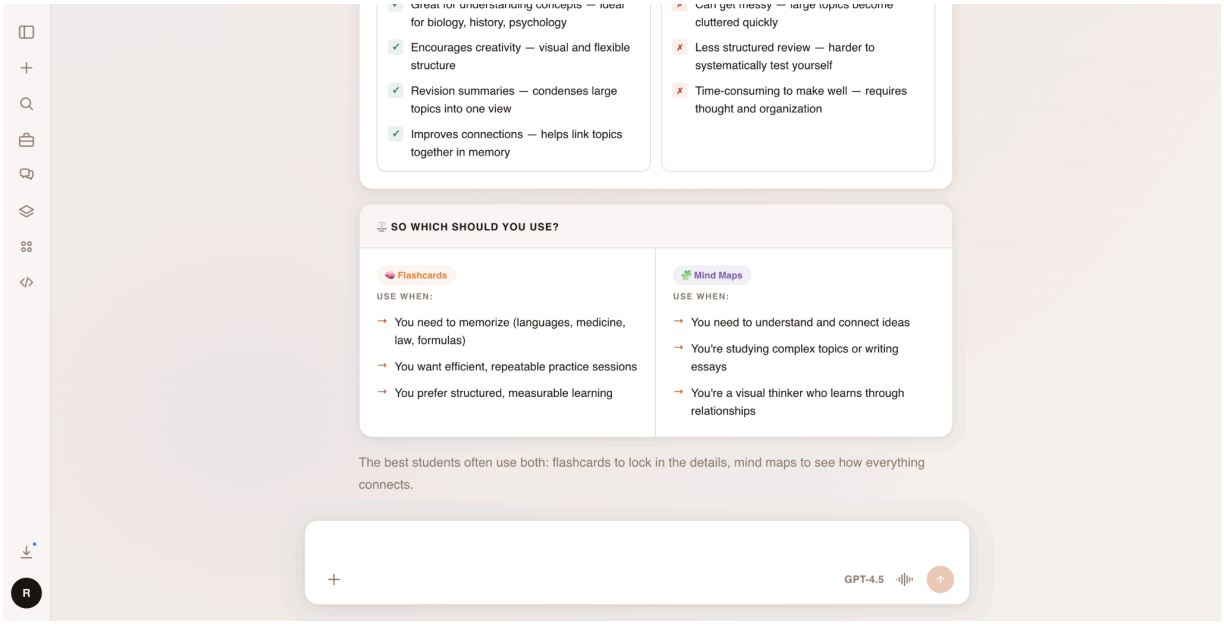
Appendix 6 Adaptive prototype of the Decision making task, Flashcards section



Appendix 7 Adaptive prototype of the Decision making task, Mind maps section



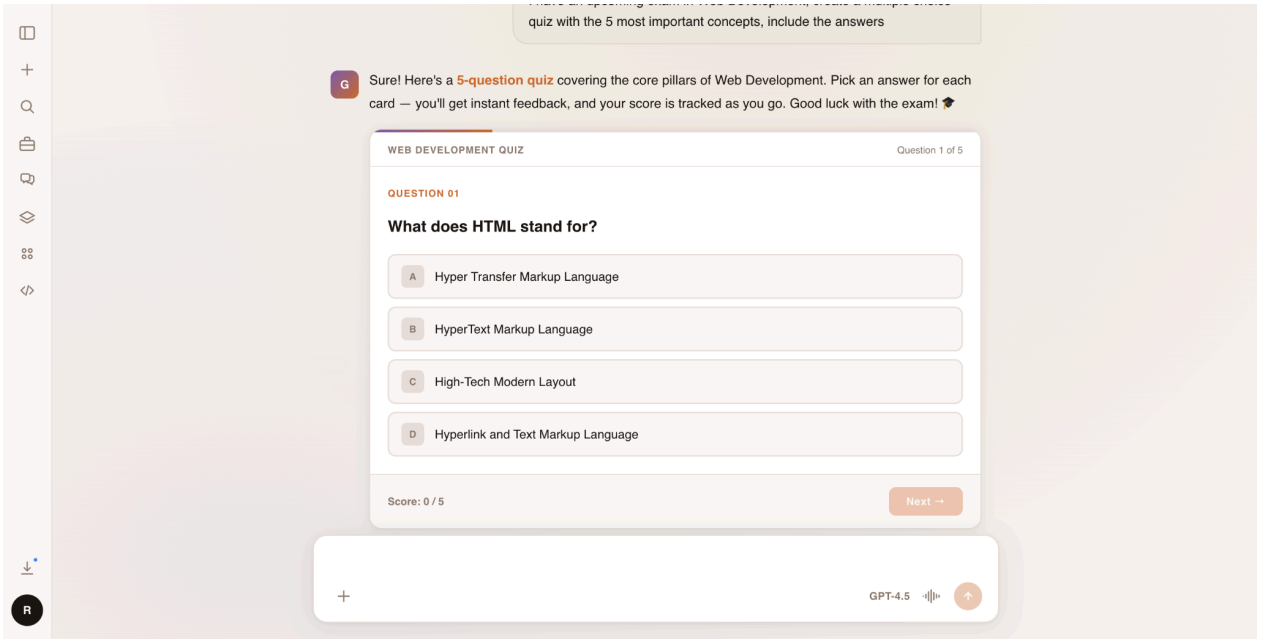
Appendix 8 Adaptive prototype of the Decision making task, conclusion section



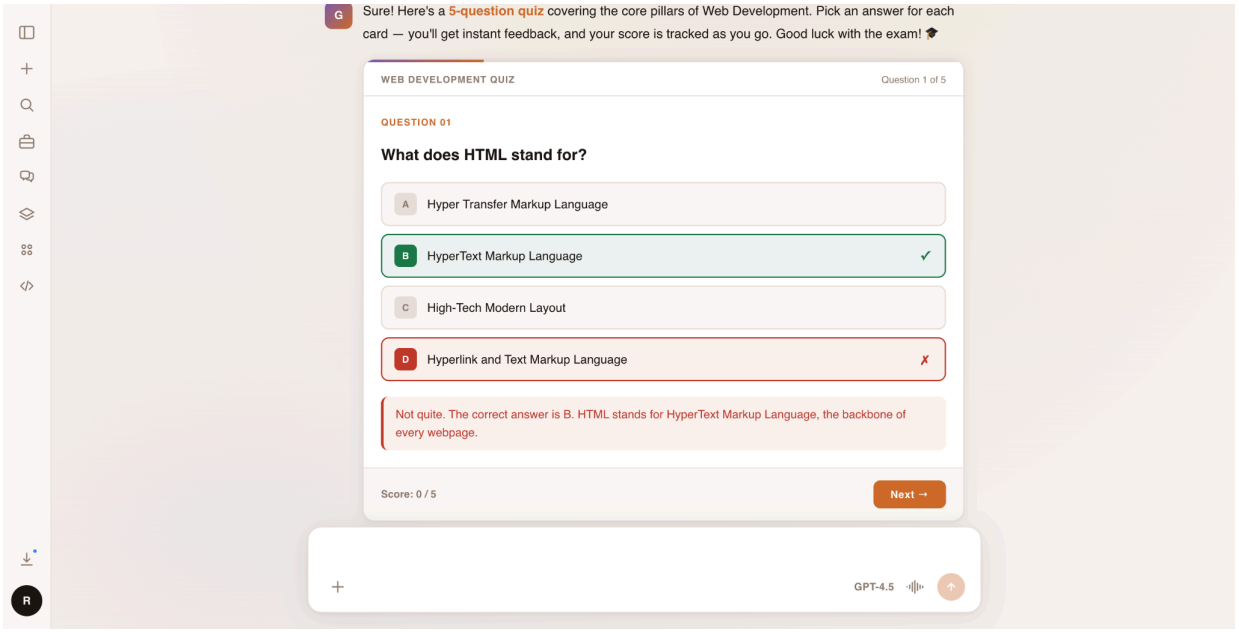
Appendix 9 Plain-text prototype of the Exam preparation task



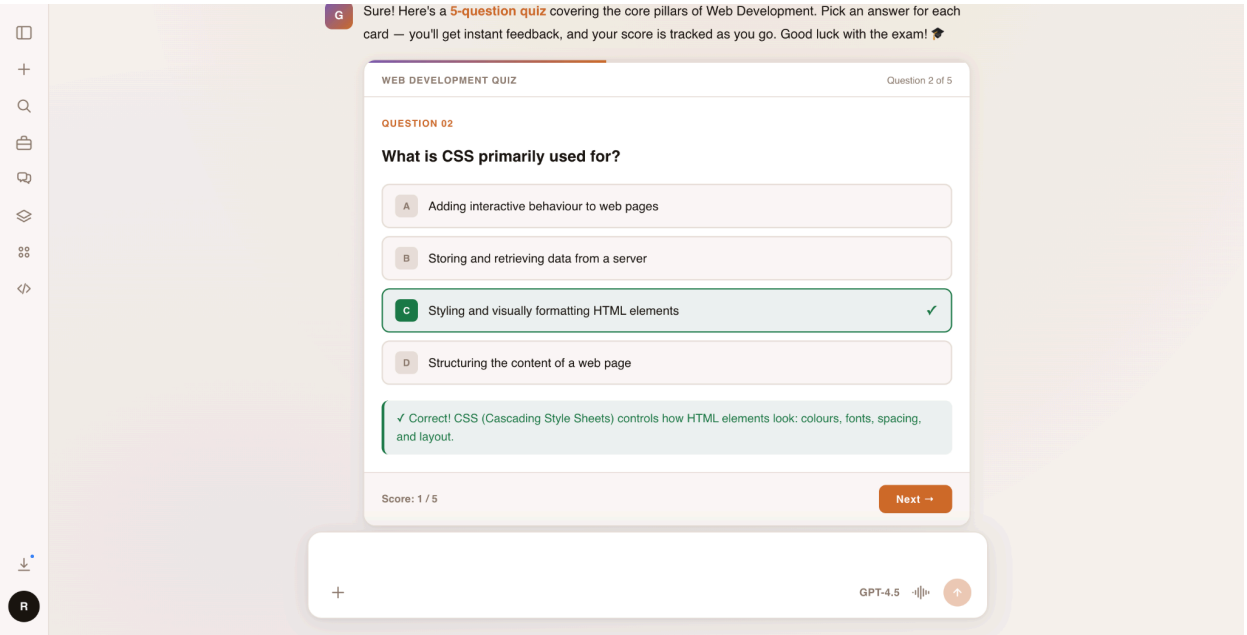
Appendix 10 Adaptive prototype of the Exam preparation task



Appendix 11 Adaptive prototype of the Exam preparation task, incorrectly selected answer result



Appendix 12 Adaptive prototype of the Exam preparation task, correctly selected answer result



Appendix 13 Researcher data entry form

Q1. Do you consent to participating in our user study and are aware that you are allowed to opt out whenever you want to? Single choice: Yes / No

Baseline questions

Q2. How often do you use AI chatbots? 5-point scale: 1 (Never) - 5 (Every day)

Q3. What types of questions do you normally ask? Multi-answer: Paraphrase / Grammar; Exam prep; Decision making; Explanation; Other (open response)

Q4. What type of AI chatbot do you use? Multi-answer: GPT; Perplexity; Claude; Grok; Gemini; Other (open response)

Decision Making - Plain-text UI

Q5. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q6. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q7. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Decision Making - Adaptive UI

Q8. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q9. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q10. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Decision Making - Task preference

Q11. Have you ever used AI for this type of task? Single choice: Yes / No / Other (open response)

Q12. Which version would you prefer to use for this type of task? Single choice: Adaptive / Plain-text / Other (open response)

Exam Preparation - Plain-text UI

Q13. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q14. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q15. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Exam Preparation - Adaptive UI

Q16. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q17. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q18. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Exam Preparation - Task preference

Q19. Have you ever used AI for this type of task? Single choice: Yes / No / Other (open response)

Q20. Which version would you prefer to use for this type of task? Single choice: Adaptive / Plain-text / Other (open response)

Grammar Check - Plain-text UI

Q21. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q22. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q23. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Grammar Check - Adaptive UI

Q24. How clearly was the information presented? 9-point scale: 1 (Very unclear) - 9 (Very clear)

Q25. Did any elements of the output draw your attention away from what mattered? Single choice: No / Other (open response)

Q26. How much mental effort did it take to extract the information? 9-point scale (PAAS): 1 (Very, very low mental effort) - 9 (Very, very high mental effort)

Grammar Check - Task preference

Q27. Have you ever used AI for this type of task? Single choice: Yes / No / Other (open response)

Q28. Which version would you prefer to use for this type of task? Single choice: Adaptive / Plain-text / Other (open response)

Closing

Q29. Any additional notes? Open response